

BEEE Unit II

Electrical Machines & Measuring Instruments

A complete last-day revision formula sheet — all formulas, principles, working, constructions, tables, flashcards & exam tips in one place.

SUBJECT
BEEE

UNIT
Unit II

PAGES
~14 Pages

FLASHCARDS
30 Q&A

 **DC Motor**

 **DC Generator**

 **Transformer**

 **Induction Motor**

 **Alternator**

 **PMMC**

 **Moving Iron**

 **Wheatstone Bridge**

SUBJECT: BASIC ELECTRICAL & ELECTRONICS ENGINEERING | UNIT II FORMULA SHEET

Construction

Stator (Stationary Part)

- **Yoke:** Outer frame, provides mechanical support & magnetic path
- **Field Poles:** Electromagnets that create main magnetic flux
- **Field Winding:** Coils wound on poles to produce flux

Rotor (Rotating Part)

- **Armature Core:** Laminated silicon steel, carries armature winding
- **Commutator:** Copper segments, converts AC → DC (in generator) / maintains rotation direction (in motor)
- **Brushes:** Carbon brushes maintain contact with commutator

Working Principle: When a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force — *Lorentz Force Law*: $F = BIL$. The armature rotates due to this force (Fleming's Left-Hand Rule gives direction of rotation).

Key Formulas

BACK EMF

$$E_b = V - I_a R_a$$

V = Supply voltage, I_a = Armature current, R_a = Armature resistance

BACK EMF (SPEED RELATION)

$$E_b = \frac{P \Phi N Z}{60 A}$$

P =poles, Φ =flux/pole, N =speed(rpm),
 Z =conductors, A =parallel paths

TORQUE EQUATION

$$T = \frac{P \Phi I_a Z}{2 \pi A}$$

or $T = 0.159 \times (PZ/A) \times \Phi \times I_a$

POWER DEVELOPED

$$P = E_b \times I_a$$

Mechanical power developed by armature

SPEED RELATION

$$N \propto E_b / \Phi$$

Speed proportional to back EMF, inversely to flux

EFFICIENCY

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

$P_{in} = V \times I_L$

Types of DC Motors

Type	Connection	Speed Characteristic	Torque	Applications
Series Motor	Field & armature in series	Variable (high at low load)	Very high starting torque	Cranes, hoists, traction, elevators
Shunt Motor	Field parallel to armature	Approximately constant	Moderate	Lathes, fans, blowers, machine tools
Compound Motor	Both series & shunt field	Adjustable	High	Presses, elevators, rolling mills

Memory Tip: Series motor — Never run at NO LOAD (speed becomes dangerously high). Shunt motor — Nearly constant speed, ideal for precision work.

⚡ DC Generator

Construction · Working Principle · EMF Equation · Types · Applications

🔧 Construction

Magnetic System

- Yoke (frame)
- Pole cores & pole shoes
- Field winding

Armature System

- Armature core (laminated)
- Armature winding
- Commutator

Commutation System

- Carbon brushes
- Brush holders
- Brush gear

Working Principle (Faraday's Law): When an armature coil rotates in a magnetic field, an EMF is induced ($e = d\Phi/dt$). The commutator converts the induced AC into unidirectional DC output. Direction given by *Fleming's Right-Hand Rule*.

📐 EMF Equation of DC Generator

$$E = \frac{P\Phi NZ}{60A}$$

Where:

P = Number of poles | Φ = Flux per pole (Wb) | N = Speed (rpm)

Z = Total conductors | A = Parallel paths

For Lap winding: $A = P$ For Wave winding: $A = 2$

TERMINAL VOLTAGE

$$V_t = E - I_a R_a$$

Generator: terminal voltage less than EMF

ARMATURE CURRENT (SHUNT)

$$I_a = I_L + I_f$$

I_L = load current, I_f = field current

EFFICIENCY

$$\eta = \frac{V_t \cdot I_L}{V_t \cdot I_L + \text{Losses}} \times 100$$

Commercial efficiency

💡 Types of DC Generators

Type	Excitation	Voltage Regulation	Applications
Series Generator	Field in series with armature	Poor (varies with load)	Boosters, arc welding
Shunt Generator	Field parallel to armature	Good (nearly constant)	Charging batteries, lighting
Compound Generator	Both series + shunt field	Excellent	Power supply, traction
Separately Excited	External DC source	Best control	Laboratory, testing

⚖️ DC Motor vs DC Generator

Feature	DC Motor	DC Generator
Energy Conversion	Electrical → Mechanical	Mechanical → Electrical
Input	Electrical supply (V, I)	Mechanical (prime mover)
Output	Mechanical (torque, speed)	Electrical (EMF, current)
Back EMF	$E_b = V - I_a R_a$	$V_t = E - I_a R_a$

Commutator role	AC → DC for armature	AC → DC output
Fleming's Rule	Left-Hand Rule	Right-Hand Rule
Applications	Drives, pumps, traction	Power generation, welding

Single Phase Transformer

Construction · Working · EMF Equation · Turns Ratio · Losses

Construction

Core Type Transformer

- Windings surround the core
- Better cooling
- Used for HV applications
- L-shaped or U-shaped laminations

Shell Type Transformer

- Core surrounds the windings
- Better mechanical protection
- Used for LV, high current
- E-I shaped laminations

Working Principle (Mutual Induction): AC supply to primary winding creates alternating magnetic flux in the core. This changing flux links with the secondary winding and induces EMF by Faraday's Law. Energy is transferred magnetically — NO electrical connection between primary & secondary.

EMF Equation & Key Formulas

Where: f = frequency (Hz), N = number of turns, Φ_m = maximum flux (Wb)

Primary: $E_1 = 4.44 f N_1 \Phi_m$ **Secondary:** $E_2 = 4.44 f N_2 \Phi_m$

$4.44 = 2\pi/\sqrt{2} = 4.4429$ (form factor for sinusoidal wave)

Turns Ratio

$$a = N_1/N_2 = E_1/E_2 = I_2/I_1$$

Voltage ratio = Turns ratio

Step-Up Condition

$$N_2 > N_1 \rightarrow E_2 > E_1$$

Turns ratio $a < 1$

Step-Down Condition

$$N_2 < N_1 \rightarrow E_2 < E_1$$

Turns ratio $a > 1$

Efficiency

$$\eta = \frac{P_{out}}{(P_{out} + P_{loss})} \times 100\%$$

Ideal transformer: $\eta = 100\%$

Max Flux

$$\Phi_m = V_1 / (4.44 \times f \times N_1)$$

Maximum flux in core

Input VA = Output VA

$$E_1 I_1 = E_2 I_2$$

Ideal transformer (no losses)

Losses in Transformer

Type	Also Called	Cause	Formula / Notes	Reduction
Iron (Core) Loss	No-load loss, Constant loss	Alternating flux in core	$P_i = P_{\text{hysteresis}} + P_{\text{eddy current}}$	CRGO steel laminations
Copper Loss	I^2R loss, Variable loss	Current through windings	$P_{cu} = I_1^2 R_1 + I_2^2 R_2$	Larger cross-section wire

Three Phase Induction Motor

Construction · Working · Slip Formula · Applications

Stator

- Outer stationary part
- Carries 3-phase winding displaced 120° apart
- Produces Rotating Magnetic Field (RMF)
- Laminated silicon steel core

Rotor

Squirrel Cage Rotor:

- Bars short-circuited by end rings
- Rugged, cheap, maintenance-free

Slip Ring (Wound) Rotor:

- 3-phase winding + external resistance
- Better starting torque control

Working Principle (Rotating Magnetic Field): 3-phase supply to stator creates a Rotating Magnetic Field (RMF) at synchronous speed N_s . This RMF cuts rotor bars → induces EMF → current flows → force on rotor → rotor starts rotating in direction of RMF (Lenz's Law). Rotor can never reach N_s (otherwise no relative motion, no induced EMF, no torque).

Key Formulas

$$s = (N_s - N) / N_s$$

Slip: s = Fractional slip (0 to 1)

N_s = Synchronous speed (rpm), N = Actual rotor speed (rpm)

At standstill: $s = 1$ At synchronous speed: $s = 0$

Typical full-load slip: 3%–8%

SYNCHRONOUS SPEED

$$N_s = 120f / P$$

f =supply frequency, P =poles

SLIP %

$$\%s = (N_s - N) / N_s \times 100$$

Percentage slip

ROTOR SPEED

$$N = N_s(1 - s)$$

Actual mechanical speed

ROTOR FREQUENCY

$$f_r = s \times f$$

Frequency of rotor induced EMF

TORQUE

$$T \propto sE_2^2R_2 / (R_2^2 + s^2X_2^2)$$

E_2 =rotor EMF, R_2 =rotor resistance

EFFICIENCY

$$\eta = (1 - s) \times 100\%$$

Approx. for ideal induction motor

Transformer vs Induction Motor

Feature	Transformer	3-Phase Induction Motor
Type	Static device	Rotating machine
Principle	Mutual induction	Rotating magnetic field
Energy conversion	Electrical → Electrical	Electrical → Mechanical
EMF equation	$E = 4.44fN\Phi$	Same (for stator/rotor)
Slip	Not applicable	$s = (N_s - N) / N_s$
Frequency of secondary	Same as primary (f)	Rotor: $f_r = sf$

Applications	Power transmission, step up/down	Pumps, fans, compressors, mills
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Alternator (Synchronous Generator)

Construction · Working · EMF Equation · Applications

Stator (Armature)

- Stationary 3-phase winding
- Distributed in slots of laminated core
- Output terminal (EMF generated here)
- Star or delta connected

Rotor (Field)

Salient Pole Rotor: Low speed (hydro), projecting poles
Cylindrical (Non-salient) Rotor: High speed (steam turbines), smooth surface, better balance at high speed

Working Principle: Rotor (field winding) is excited by DC supply → creates magnetic flux → rotor is driven by prime mover → rotating magnetic field cuts stator conductors → EMF induced by Faraday's Law → 3-phase AC output at stator terminals.

EMF Equation of Alternator

$$E_{rms} = 4.44 K_w f N \Phi_m$$

Where: K_w = winding factor (0.85–0.95), f = frequency (Hz)
 N = turns per phase, Φ_m = maximum flux (Wb)
For concentrated winding: $K_w = 1 \rightarrow E = 4.44fN\Phi_m$

FREQUENCY

$$f = PN_s / 120$$

P=poles, N_s =rotor speed(rpm)

LINE VOLTAGE (STAR)

$$V_L = \sqrt{3} \times V_{Ph}$$

Star connected stator

VOLTAGE REGULATION

$$VR = (E_0 - V) / V \times 100\%$$

E_0 =no-load EMF, V=rated voltage

Parameter	DC Generator	Alternator
Output	DC	AC (3-phase)
Armature	Rotating	Stationary (stator)
Field	Stationary	Rotating (rotor)
Commutator	Required	Not required
EMF generation	Faraday + commutation	Faraday's Law (direct AC)
Applications	Electroplating, DC drives	Power stations, turbines

All Machine Applications Summary

Machine	Key Applications
DC Motor (Series)	Cranes, hoists, electric locomotives, traction
DC Motor (Shunt)	Lathes, grinders, fans, blowers, machine tools
DC Generator	Battery charging, electroplating, arc welding
Transformer	Power transmission, distribution, isolation, voltage conversion
Induction Motor	Pumps, fans, compressors, conveyor belts, mills (most used motor)

Alternator	Power plants (hydro, thermal, nuclear), emergency generators
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Measuring Instruments — PMMC

Permanent Magnet Moving Coil Instrument

Construction of PMMC

Permanent Magnet System

- **Permanent Magnet:** Alnico or ferrite, provides strong steady field
- **Pole Pieces:** Shaped to give uniform radial field
- **Soft Iron Core:** Cylindrical, concentrates flux

Moving System

- **Rectangular Coil:** Wound on aluminum former
- **Pivots + Jewel Bearings:** Supports moving coil
- **Hair Springs:** Provide controlling torque, conduct current
- **Pointer + Scale:** Deflects over graduated scale

Working Principle: Current through the moving coil in the field of permanent magnet → Deflecting torque ($T_d = BANl$) → Coil deflects → Controlling torque from hair springs ($T_c = k\theta$) → At equilibrium $T_d = T_c$ → Deflection $\theta \propto I$ (linear scale).

DEFLECTING TORQUE

$$T_d = BANl$$

B=flux density, A=coil area, N=turns,
I=current

CONTROLLING TORQUE

$$T_c = k\theta$$

k=spring constant, θ =deflection angle

AT EQUILIBRIUM

$$\theta = BANl/k$$

$\theta \propto I \rightarrow$ uniform (linear) scale

Advantages & Disadvantages

Advantages

- Uniform / linear scale
- High sensitivity & accuracy
- Low power consumption
- No hysteresis error
- Good damping (eddy current in Al former)
- Not affected by external stray fields

Disadvantages

- Works ONLY on DC (not AC)
- Fragile — delicate construction
- Costly (permanent magnets)
- Errors due to temperature
- Ageing of permanent magnet
- Cannot withstand overload

Applications of PMMC

DC Voltmeters

DC Ammeters

Galvanometers

Ohmmeter (with battery)

Multimeters (with rectifier for AC)

Panel meters

Signal generators

Key Point: PMMC can measure AC if a rectifier (diode bridge) is used. The rectifier converts AC to DC before it reaches the PMMC movement.

Extension of PMMC Range

AMMETER (SHUNT RESISTOR)

$$R_{sh} = I_m R_m / (I - I_m)$$

I_m =FSD current, R_m =meter resistance, I=total current

VOLTMETER (SERIES RESISTOR)

$$R_{se} = V/I_m - R_m$$

V=voltage to measure, series multiplier

Types of Moving Iron Instruments

Attraction Type

- Single fixed coil + soft iron vane
- Iron vane attracted into the coil on current flow
- Simpler, less accurate
- Suitable for low frequencies

Repulsion Type (Common)

- Fixed iron vane + movable iron vane inside coil
- Both vanes magnetized in same direction → repulsion → deflection
- More uniform scale, widely used
- Better accuracy

Working Principle (MI): Current flows through fixed coil → magnetizes both iron vanes with same polarity → like poles repel → movable vane deflects → deflection proportional to I^2 . Since $T_d \propto I^2$, scale is non-uniform (cramped at low end). Works on BOTH AC and DC.

DEFLECTING TORQUE

$T_d = \frac{1}{2} I^2 (dL/d\theta)$

L=coil inductance, I=current

CONTROLLING TORQUE

$T_c = k\theta$

Spring control (or gravity)

DEFLECTION

$\theta \propto I^2$

Non-linear (squared) scale

PMMC vs Moving Iron Instruments

Feature	PMMC	Moving Iron (MI)
Working current	DC only	AC and DC both
Scale	Uniform / Linear ($\theta \propto I$)	Non-uniform ($\theta \propto I^2$)
Torque source	Permanent magnet + coil	Magnetized iron vanes
Sensitivity	Very high	Lower
Accuracy	High (Class 0.1, 0.25)	Moderate (Class 1.5, 2.5)
Frequency response	DC only	Up to 125 Hz
Damping	Eddy current (Al former)	Air friction damping
Cost	High	Low
Hysteresis error	None	Present in iron
Applications	DC lab instruments, multimeters	AC panel meters, switchboards



Wheatstone Bridge

NETWORK

Circuit Concept: Four resistors R_1, R_2, R_3, R_4 arranged in a diamond (bridge) shape. A battery is connected across one diagonal and a galvanometer (G) across the other. Used for precise measurement of unknown resistance.

$R_1/R_2 = R_3/R_4$

Balance Condition (Null Condition): No current through galvanometer ($I_G = 0$)

Cross-multiplication form: $R_1 \times R_4 = R_2 \times R_3$

Unknown: $R_x = R_2 \times R_3 / R_1$ (if $R_4 = R_x$)

UNKNOWN RESISTANCE

$R_x = R_s \times (P/Q)$

R_s =standard arm, P & Q = ratio arms

SENSITIVITY

$S = \Delta I_G / \Delta R$

Change in galvanometer current per unit resistance change

Arm	Designation	Role
P (R_1)	Ratio Arm	Known, variable resistor
Q (R_2)	Ratio Arm	Known, variable resistor
R (R_3)	Standard Arm	Known standard resistance
S (R_4)	Unknown Arm	Unknown resistance to measure

✓ Applications

- Measurement of resistance (1Ω to 1MΩ)
- Temperature measurement (RTD)
- Strain gauge measurements
- Fault location in cables
- Inductance & capacitance (modified bridges)

⚠ Limitations

- Not suitable for very low (<1Ω) or very high resistance (>1MΩ)
- Heating effect at high currents (error)
- Contact and lead resistance cause errors
- Balance condition requires careful adjustment

**Q:** Working principle of DC Motor?**A:** Force on current-carrying conductor in a magnetic field (Lorentz force). Direction by **Fleming's Left-Hand Rule**.**Q:** Torque equation of DC Motor?

A: $T = P\Phi I_a Z / 2\pi A$

$T \propto \Phi \times I_a$

Q: EMF equation of Transformer?

A: $E = 4.44 f N \Phi_m$

4.44 is the form factor ($2\pi/\sqrt{2}$)**Q:** Slip in Induction Motor?

A: $s = (N_s - N) / N_s$

At standstill $s=1$; at synchronous speed $s=0$ **Q:** Wheatstone bridge balance condition?

A: $R_1/R_2 = R_3/R_4$

or $R_1 R_4 = R_2 R_3$

Q: Why PMMC can't measure AC directly?**A:** AC produces alternating torque $\rightarrow$ average = 0 $\rightarrow$ pointer stays at zero. Needs rectifier for AC measurement.**Q:** Why MI scale is non-uniform?**A:** Because $\theta \propto I^2$ — scale is cramped (compressed) at low values and expanded at higher values.**Q:** Why can't an IM reach synchronous speed?**A:** If $N = N_s$, slip = 0 $\rightarrow$ no relative motion $\rightarrow$ no induced EMF $\rightarrow$ no rotor current $\rightarrow$ no torque $\rightarrow$ motor stops. Hence N always $< N_s$.**Q:** Back EMF formula of DC Motor?

A: $E_b = V - I_a R_a$

Also: $E_b = P\Phi NZ/60A$

Q: EMF equation of DC Generator?

A: $E = P\Phi NZ / 60A$

 P =poles, Φ =flux, N =rpm, Z =conductors, A =paths**Q:** Turns ratio of a Transformer?

A: $a = N_1/N_2 = E_1/E_2 = I_2/I_1$

Q: Synchronous speed formula?

A: $N_s = 120f / P$

 f =frequency (Hz), P =number of poles**Q:** PMMC deflection equation?

A: $\theta = BANl / k$

 $\theta \propto I$ — uniform (linear) scale**Q:** Working principle of Moving Iron instrument?**A:** Current through coil magnetizes two iron vanes with same polarity $\rightarrow$ repulsion $\rightarrow$ deflection. $T_d \propto I^2$ **Q:** Rotor frequency of Induction Motor?

A: $f_r = s \times f$

 s =slip, f =supply frequency

**Q: Working principle of DC Generator?**

A: Rotating armature in magnetic field → EMF induced (Faraday's Law). Direction by **Fleming's Right-Hand Rule**.

Q: What is Back EMF? Why is it important?

A: EMF induced in motor armature opposing supply voltage. It limits armature current: $I_a = (V - E_b)/R_a$. Higher speed → higher E_b → lower I_a .

Q: Series motor: Why never run on no load?

A: At no load, very little current → very low flux → $N \propto 1/\Phi$ → speed becomes dangerously high → motor can be damaged.

Q: Transformer losses: types?

A: Iron (Core) loss = constant (hysteresis + eddy current). Copper loss = variable (I^2R). No moving parts → no friction losses.

Q: Damping in PMMC?

A: Eddy current damping: Al former (coil former) moves in permanent magnet field → eddy currents induced → opposing force → smooth pointer deflection.

Q: Alternator EMF equation?

A: $E = 4.44 K_w f N \Phi_m$
 K_w = winding factor (accounts for distributed winding)

Q: What is commutator? Its role?

A: Cylindrical device of copper segments. In DC motor: converts DC supply to AC in armature. In DC generator: converts induced AC to DC output.

Q: PMMC shunt resistor (ammeter extension)?

A: $R_{sh} = I_m R_m / (I - I_m)$
 I_m = FSD current, I = required range

Q: Working principle of Alternator?

A: Rotating magnetic field (rotor) → cuts stationary stator conductors → induces 3-phase AC EMF (Faraday's Law).

Q: What is significance of 4.44 in transformer EMF?

A: $4.44 = 2\pi/\sqrt{2} = \pi\sqrt{2}$
 It is the product of form factor (1.11) and 2π (=6.28). Applies only for sinusoidal waveforms.

Q: Lap winding vs Wave winding?

A: Lap: $A = P$ (more parallel paths, more current, low voltage).
 Wave: $A = 2$ (less paths, high voltage, low current).

Q: What is rotating magnetic field (RMF)?

A: 3-phase supply to stator windings (displaced 120°) creates a resultant magnetic flux that rotates at synchronous speed $N_s = 120f/P$.

Q: Damping in Moving Iron instrument?

A: Air friction damping: aluminum vane moves inside air chamber → air resistance opposes motion → damping achieved.

Q: Efficiency of induction motor (approximate)?

A: $\eta \approx (1 - s) \times 100\%$
 Lower slip → higher efficiency. Typical: 85–95%

Q: Wheatstone bridge — when is it most sensitive?

A: When all four arms have equal resistance ($R_1=R_2=R_3=R_4$).
 Bridge is most sensitive near the balance condition.



Exam Tips & Common Mistakes

MUST READ

- 1 **Motor vs Generator Confusion:** Motor = Electrical $\rightarrow$ Mechanical (Fleming's Left-Hand). Generator = Mechanical $\rightarrow$ Electrical (Fleming's Right-Hand). Memory: "*Mo-Left, Gen-Right*" or "*Motors Live, Generators Remain*".
- 2 **Forgetting the 4.44 factor in EMF equation:** Always use $E = 4.44 f N \Phi_m$ — NOT $2\pi f N \Phi$ (that is the peak value divided by $\sqrt{2}$ form). $4.44 = 2\pi \times (1/\sqrt{2})$ = rms value formula constant. It applies to sinusoidal supply ONLY.
- 3 **Slip Calculation Errors:** Slip = $s = (N_s - N) / N_s$ — denominator is always N_s (synchronous speed), NOT actual speed N . Also: rotor speed $N = N_s(1-s)$, NOT $N_s - s$.
- 4 **Mixing PMMC and MI Characteristics:** PMMC = DC only, uniform scale ($\theta \propto I$). MI = AC & DC, non-uniform scale ($\theta \propto I^2$). Never say MI has uniform scale or PMMC works on AC.
- 5 **Ignoring Units and Symbols:** Φ = flux in Webers (Wb), B = flux density in Tesla (T), N = turns (no unit), f = Hz, T = Torque in N·m, E = EMF in Volts (V). Always check units in numerical problems.
- 6 **Incorrect Wheatstone Bridge Balance:** Balance condition is $R_1/R_2 = R_3/R_4$ or equivalently $R_1R_4 = R_2R_3$. Do NOT mix up the arms — opposite arms must multiply equally.
- 7 **Back EMF confusion:** Back EMF opposes supply. In motor: $E_b = V - I_a R_a$, so $E_b < V$. In generator: $V_t = E - I_a R_a$, so terminal voltage $<$ generated EMF.
- 8 **Parallel paths A in EMF equation:** For Lap winding: $A = P$ (number of poles). For Wave winding: $A = 2$. Always check winding type in numerical problems before substituting A .
- 9 **Transformer — No rotating parts:** Since transformer is static, it has NO friction/windage losses. Only Iron loss (constant) + Copper loss (variable I^2R). Efficiency can exceed 95–99%.
- 10 **Series DC Motor — Never no load:** Series motor field flux $\propto I_a$. At no load, $I_a \rightarrow 0 \rightarrow \Phi \rightarrow 0 \rightarrow N \rightarrow \infty$. This is dangerous. Always load the series motor before starting.

High-Risk Exam Traps:

- Using rms formula ($2\pi f N \Phi$) instead of $4.44 f N \Phi$ for transformer EMF
- Writing $\theta \propto I$ for MI instruments (correct: $\theta \propto I^2$)
- Saying $N_s - N$ in denominator for slip (denominator must be N_s)
- Mixing Fleming's Left (motor) and Right (generator) hand rules
- Saying transformer works on self-induction (it works on MUTUAL induction)
- Forgetting K_w (winding factor) in alternator EMF equation



⚙️ DC MOTOR

Back EMF

$$E_b = V - I_a R_a$$

Back EMF (speed)

$$E_b = P\Phi NZ/60A$$

Torque

$$T = P\Phi I_a Z/2\pi A$$

Speed $\propto$

$$N \propto E_b/\Phi$$

Power developed

$$P = E_b \times I_a$$

⚡ DC GENERATOR

Generated EMF

$$E = P\Phi NZ/60A$$

Terminal Voltage

$$V_t = E - I_a R_a$$

Lap winding

$$A = P$$

Wave winding

$$A = 2$$

Efficiency

$$\eta = P_{out}/P_{in} \times 100\%$$

📄 TRANSFORMER

EMF Equation

$$E = 4.44fN\Phi_m$$

Turns Ratio

$$a = N_1/N_2 = E_1/E_2$$

Current Ratio

$$I_1/I_2 = N_2/N_1$$

Max Flux

$$\Phi_m = V_1/(4.44fN_1)$$

VA balance

$$E_1 I_1 = E_2 I_2$$

🌀 INDUCTION MOTOR

Synchronous Speed

$$N_s = 120f/P$$

Slip

$$s = (N_s - N)/N_s$$

Rotor Speed

$$N = N_s(1-s)$$

Rotor Frequency

$$f_r = s \times f$$

Efficiency (approx)

$$\eta \approx (1-s) \times 100\%$$

🔄 ALTERNATOR

EMF Equation

$$E = 4.44K_w f N \Phi_m$$

Frequency

$$f = PN_s/120$$

Line Voltage (Star)

$$V_L = \sqrt{3} V_{ph}$$

VR %

$$(E_0 - V)/V \times 100$$

🔧 PMMC INSTRUMENT

Deflecting Torque

$$T_d = BAN I$$

Deflection

$$\theta = BAN I/k$$

Scale type

$$\theta \propto I \text{ (uniform)}$$

Shunt (Ammeter)

$$R_{sh} = I_m R_m / (I - I_m)$$

Series (Voltmeter)

$$R_{se} = V/I_m - R_m$$

📊 MOVING IRON

Deflecting Torque

$$T_d = \frac{1}{2} I^2 (dL/d\theta)$$

Deflection

$$\theta \propto I^2 \text{ (non-uniform)}$$

Works on

AC & DC both

Damping

Air friction

🔵 WHEATSTONE BRIDGE

Balance Condition

$$R_1/R_2 = R_3/R_4$$

Cross Form

$$R_1 R_4 = R_2 R_3$$

Unknown R

$$R_x = R_s \times P/Q$$

Null condition

$$I_g = 0$$

📋 KEY CONSTANTS

Form factor

$$4.44 = 2\pi/\sqrt{2}$$

π value

$$\pi \approx 3.14159$$

$\sqrt{3}$ value

$$\sqrt{3} \approx 1.732$$

$\sqrt{2}$ value

$$\sqrt{2} \approx 1.414$$

50 Hz $\rightarrow$ N_s (2 pole)

$$3000 \text{ rpm}$$

⚡ Key Principles (One-Line Revision)

Machine / Device	Principle	Law / Rule
DC Motor	Current-carrying conductor in magnetic field $\rightarrow$ Force	Lorentz Force, Fleming's Left-Hand
DC Generator	Moving conductor in magnetic field $\rightarrow$ EMF induced	Faraday's Law, Fleming's Right-Hand
Transformer	Changing flux in primary $\rightarrow$ EMF in secondary	Faraday's Law, Mutual Induction

Induction Motor	RMF → induced rotor current → force → rotation	Faraday's Law, Lenz's Law
Alternator	Rotating field → stationary armature EMF induced	Faraday's Law
PMMC	Current in coil + permanent field → torque → deflection	Lorentz Force, $\theta \propto I$
Moving Iron	Current → iron magnetized → repulsion → deflection	Magnetic repulsion, $\theta \propto I^2$
Wheatstone Bridge	Null deflection of galvanometer at balance	Kirchhoff's Voltage Law



► ALL MACHINES — TYPES, PRINCIPLE & APPLICATIONS

Machine	Type	Energy Conversion	Key Principle	Common Applications
DC Motor	Series	Elec → Mech	Lorentz Force	Cranes, hoists, traction
	Shunt			Lathes, fans, machine tools
	Compound			Presses, elevators
DC Generator	Series	Mech → Elec	Faraday's Law	Boosters, arc lighting
	Shunt			Battery charging, lighting
	Compound			Power supply, traction
Transformer	Core / Shell	Elec → Elec	Mutual Induction	Power grids, distribution, isolation
Induction Motor	Squirrel Cage	Elec → Mech	RMF + Lenz's Law	Pumps, fans, compressors
	Slip Ring			Hoists, cranes (variable speed)
Alternator	Salient / Cylindrical	Mech → Elec (AC)	Faraday's Law (rotating field)	Power plants, ships, emergency gen

► INSTRUMENTS COMPARISON SUMMARY

Feature	PMMC	Moving Iron	Wheatstone Bridge
Quantity measured	V, I (DC)	V, I (AC & DC)	Resistance
Scale	Uniform (linear)	Non-uniform (squared)	Not applicable (null)
Accuracy	High (0.1–0.25 class)	Moderate (1.5–2.5)	Very High
Sensitivity	Very high	Low	Depends on G
Frequency	DC only	DC and AC (up to 125Hz)	DC (basic bridge)
Key formula	$\theta = BANIk$	$T_d = \frac{1}{2}I^2(dL/d\theta)$	$R_1/R_2 = R_3/R_4$

► ALL KEY FORMULAS SUMMARY

#	Formula	Quantity	Machine / Device
1	$E = P\Phi NZ / 60A$	Generated EMF	DC Generator / DC Motor
2	$E_b = V - I_a R_a$	Back EMF	DC Motor
3	$T = P\Phi I_a Z / 2\pi A$	Torque	DC Motor
4	$E = 4.44 f N \Phi_m$	EMF	Transformer
5	$N_1/N_2 = E_1/E_2 = I_2/I_1$	Turns Ratio	Transformer
6	$N_s = 120f / P$	Sync. Speed	Induction Motor

7	$s = (N_s - N) / N_s$	Slip	Induction Motor
8	$f_r = s \times f$	Rotor Frequency	Induction Motor
9	$E = 4.44 K_w f N \Phi_m$	EMF	Alternator
10	$f = PN_s / 120$	Frequency	Alternator
11	$\theta = BANl / k$	Deflection	PMMC
12	$T_d = \frac{1}{2} I^2 (dL/d\theta)$	Torque	Moving Iron
13	$R_1 R_4 = R_2 R_3$	Balance Condition	Wheatstone Bridge
14	$R_{sh} = I_m R_m / (I - I_m)$	Shunt Resistance	PMMC Ammeter
15	$R_{se} = V / I_m - R_m$	Series Multiplier	PMMC Voltmeter

Last Minute Checklist:

- ✓ Know EMF equation for each machine (different constants/parameters)
- ✓ Know turns ratio and what it implies for step-up vs step-down
- ✓ Know slip formula and boundary values (0 to 1)
- ✓ Know difference between PMMC and MI (scale, frequency, accuracy)
- ✓ Know Wheatstone bridge balance condition and how to find unknown R
- ✓ Know Fleming's Left-Hand (motor) and Right-Hand (generator) rules
- ✓ Always check winding type (lap/wave) before substituting A in formulas